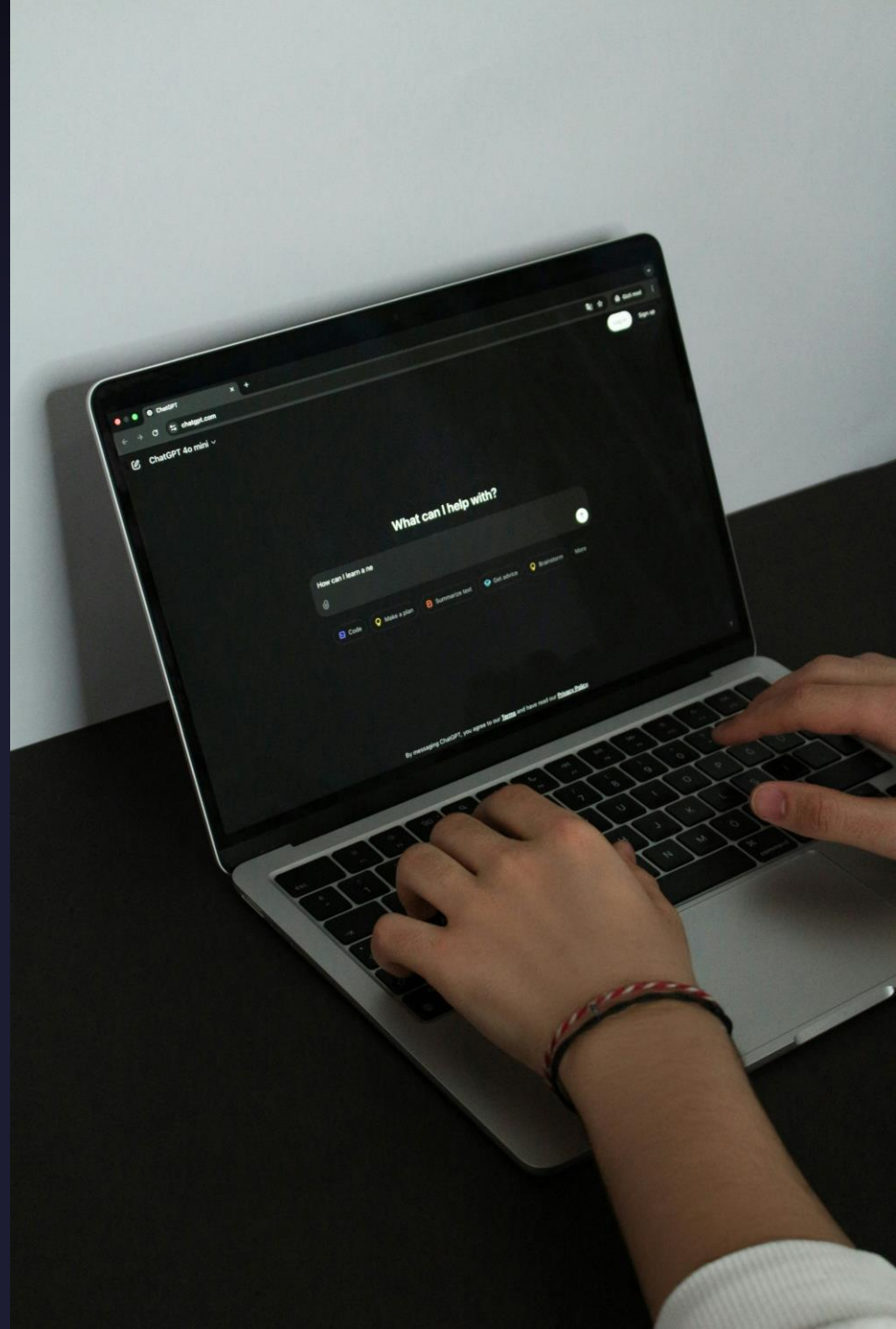
GENERATIVE AI IMMERSION



PROMPT ENGINEERING

Module 02

AGENDA

- The Paradigm Shift: Search vs Synthesis
- The Era of Generative AI
- Defining Prompt Engineering
- Prompt Framework
- Types of Prompts
- Future Outlook
- Assignments
- Next Steps

THE PARADIGM SHIFT: SEARCH VS. SYNTHESIS

Search engines such as Google search **index** and **rank** websites. When we perform a search, our keywords are used to find the websites (or web pages) with best matches, and the results are returned in a ranked order, with the closest match first.

The Traditional Hurdle: Classic search engines requires users to act as their own data aggregators (collectors), manually vetting and synthesizing information from ranked list of URLs.

The LLM Breakthrough: Generative AI bypasses the "link-clicking" phase by providing direct, context-aware answers to complex queries.

Strategic Value: By reducing the cognitive load of information retrieval, LLMs transform search from a **multi-step research task** into an **instantaneous knowledge service.**

Popular search engines

THE PARADIGM SHIFT: SEARCH VS. SYNTHESIS

Feature	Traditional Search (Retrieval)	Generative AI (Synthesis)
Mechanism	Indexing, ranking, and keyword matching.	Deep learning and contextual understanding.
User Effort	Manual Synthesis: Browsing multiple sites to piece together an answer.	Automated Synthesis: Receiving a direct, consolidated response.
Outcome	List of relevant sources (high friction).	Actionable information (low friction).
Value Prop	Comprehensive discovery.	Efficiency and "Mental Bandwidth" recovery.

THE ERA OF GENERATIVE AI

- **Non-Algorithmic Search**: Unlike Google, LLMs do not "query" a live database at inference time. They are parametric, meaning knowledge is stored within the model's weight.
- **The Pre-Training Advantage**: Models are pre-trained on petabytes of diverse data (web, research, code). In the training stage, the weights of the model's billions of connections are calculated.
- **Example**: During training, models map "Business Strategy" and "Financial Services" as separate conceptual clusters. But when prompted for "Fintech Strategy", the model performs **Cross-Domain Generalization**.
- **Combinatorial Intelligence**: It identifies the intersection of these latent spaces to generate a response that didn't exist as a single document in its training set.

Analogy for the class: It's not a digital library; it's a digital brain that has read the library and can now improvise a speech on any two topics combined.

THE ERA OF GENERATIVE AI

The Stochastic Nature of LLMs: Because LLMs are probabilistic, they are prone to systemic risks such as:

- **Hallucinations:** Confidently stating false information.
- **Degenerate Output:** Generic, repetitive, or "safe" but useless responses.
- **Format Drift:** Failing to adhere to specific schemas (JSON, Markdown, etc.).

Therefore, to harness the full potential of LLMs, a new field of **Prompt Engineering** was developed, which enables LLMs to understand intent, follow instructions, and generate desired outputs. As we increasingly interact with AI in various applications, prompt engineering plays a vital role in ensuring accurate, relevant, and safe interactions.

WHAT IS A PROMPT?

In the context of AI, a prompt is the **input** you provide to the model to generate/obtain a specific response.

This can take various forms, ranging from *simple questions* or *keywords* to *complex instructions*, code snippets, or even creative writing samples. The effectiveness of your prompt directly influences the quality and relevance of the AI's output.

Every question is a prompt, but not every prompt is a question.

WHAT IS PROMPT ENGINEERING?

Prompt Engineering (PE) is the art and science of designing and optimizing prompts to guide AI models, particularly LLMs, towards generating the desired responses. By carefully crafting prompts, you provide the model with context (background), instructions, and examples that help it understand your intent and respond in a meaningful way. Think of PE as providing a roadmap for the AI, steering it towards the specific output you have in mind.

- Effective prompts reduce the probability space of the AI model, which prevents it from generating default output.
- Prompt Engineering helps users to mitigate the systemic risks of using LLMs such as hallucinations, degenerate output, and format drift.

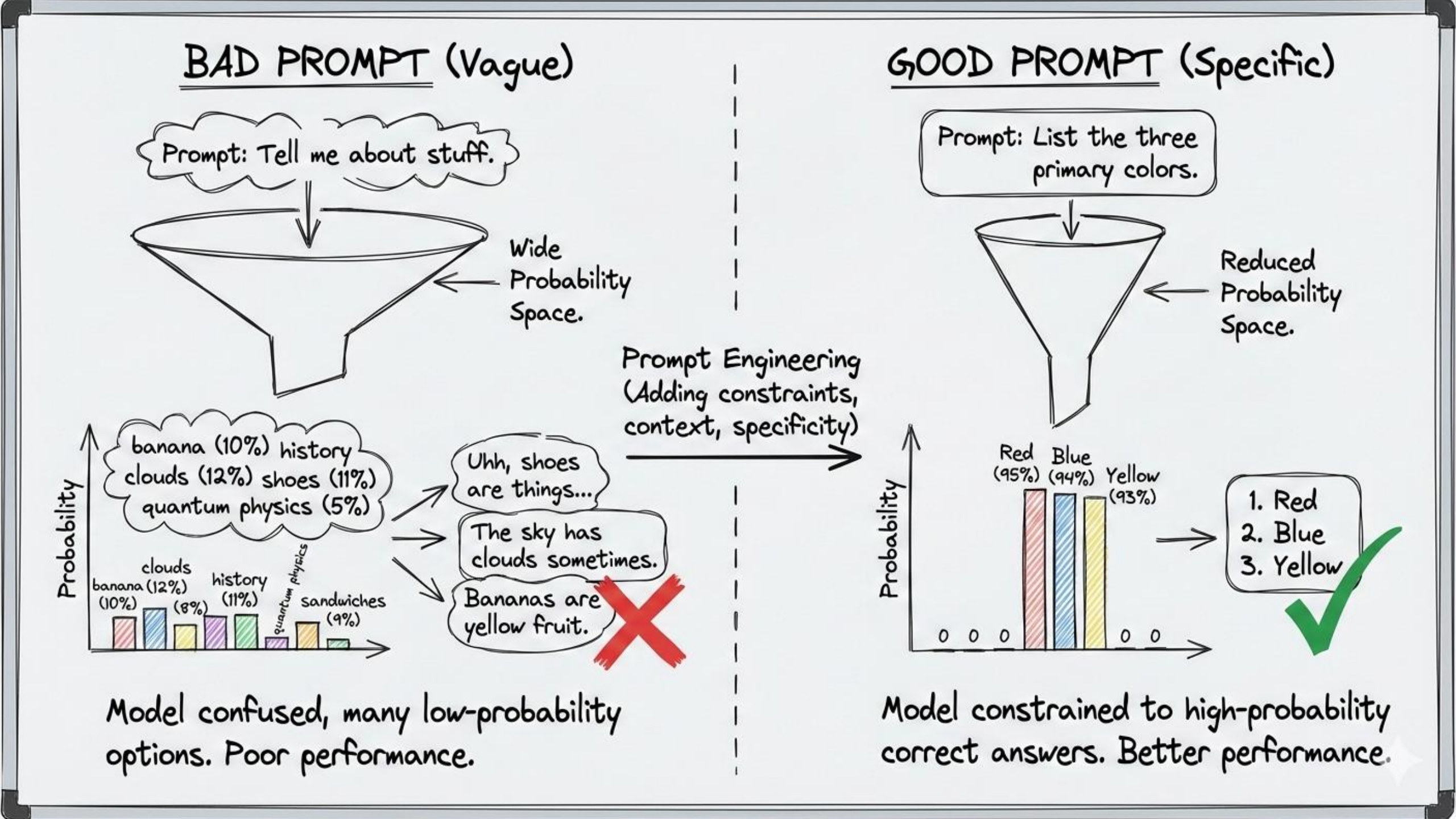
QUESTION VS. PROMPT?

A **question** is a narrow request for information, clarification, or a specific answer, typically expecting a concise and factual response (for example, a definition, explanation, or yes/no). A **prompt**, by contrast, is a structured instruction designed to guide a system—often an AI—on *how* to think, what context to consider, what constraints to follow, and what form the output should take. In short, a question asks *what* you want to know, while a prompt specifies *what to do*, *how to do it*, and *what the result should look like*.

BENEFITS OF PROMPT ENGINEERING

Effective prompt engineering offers benefits, enhancing the capabilities and usability of AI models.

1. Improved model performance.
2. Reduced bias and harmful responses.
3. Increased control and predictability.
4. Enhanced user experience.
5. Efficient use of energy and resources.


BAD PROMPT (Vague)
Prompt: Tell me about stuff.
Wide Probability Space.
banana (10%) history clouds (12%) shoes (11%) quantum physics (5%)
Probability
banana (10%)
clouds (12%)
(8%)
history (11%)
quantum physics
sandwiches (9%)
Uhh, shoes are things...
The sky has clouds sometimes.
Bananas are yellow fruit.
Model confused, many low-probability options. Poor performance.
Prompt Engineering (Adding constraints, context, specificity)
GOOD PROMPT (Specific)
Prompt: List the three primary colors.
Reduced Probability Space.
Probability
Red (95%)
Blue (94%)
Yellow (93%)
1. Red
2. Blue
3. Yellow
Model constrained to high-probability correct answers. Better performance.

PROMPT ENGINEERING IS AN ESSENTIAL SKILL

Top five skills required to reap the benefits of Gen AI			
Leadership and upper management	**Middle management**	**Frontline management**	**Entry-level employees**
Critical thinking	Risk management	Data management	Data management
Creative thinking	Data management	Prompt engineering	Prompt engineering
Data management	Ethical judgement	Technological literacy	Technological literacy
Risk management	Critical thinking	Risk management	Risk management
Ethical judgement	Change management	Emotional intelligence	Problem solving

PARTS OF A PROMPT

The four main areas to consider when writing an effective prompt are:

- Persona
- Task
- Context
- Format

> You are a program manager in [industry]. Draft an executive summary email to [persona] based on [details about relevant program docs]. Limit to bullet points.

You don't need to use all four in every prompt, but using a few will help! Always remember to include a verb or command as part of your task; this is the most important component of a prompt.

FRAMEWORK OF A GREAT PROMPT

1. **Persona**: You define a specific role or expert identity for the AI to adopt because it narrows the model's internal probability space toward a specific tone, vocabulary, and depth of knowledge.

2. **Task**: You clearly state the primary objective or action the AI must perform so that the model understands exactly what "success" looks like for the generation.

3. **Context**: You provide background information, data, or the "why" behind the request to ground the AI's logic in your specific situation rather than a generic average.

4. **Constraints**: You establish boundaries, rules, or "no-go" zones to prevent hallucinations, irrelevant tangents, or inappropriate formatting.

5. **Output Format**: You specify the structure of the final response (e.g., Table, JSON, Bullet points) to ensure the data is immediately actionable and fits your workflow.

EXAMPLE OF A GREAT PROMPT

"[Persona] Act as a Senior Investment Analyst and Luxury Tourism Consultant specializing in the East African market. [Task] Draft a professional investment pitch for a new 'Eco-Photography Lodge' located in the Simien Mountains. [Context] With the 2026 expansion of the 'Dine for Ethiopia' initiative and increased interest from international hobbyist photographers, we need to highlight the gap in high-end lodging that caters specifically to high-ISO night sky photography and endemic wildlife. [Constraints] Do not include hunting-related activities, keep the tone formal and persuasive, and ensure the word count is under 250 words. [Output Format] Provide the response in a professional memo format with three distinct headings: Market Opportunity, Strategic Value, and Infrastructure Requirements."

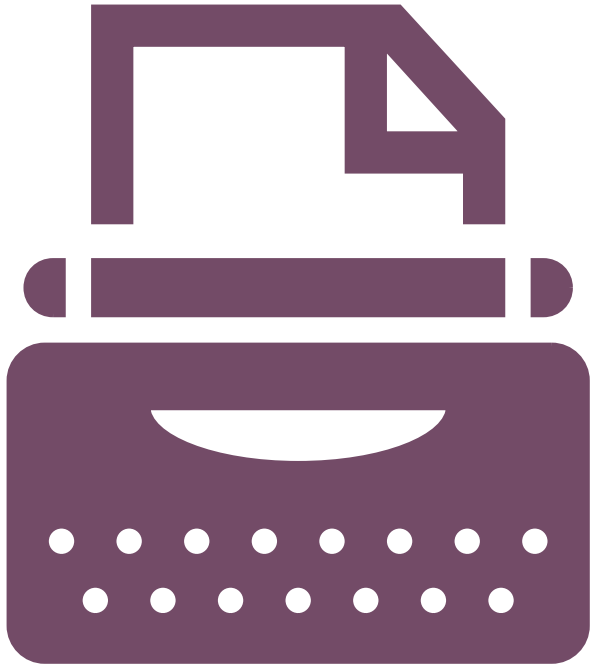
CONTEXT: THE MOST IMPORTANT VARIABLE

1. **The Default State**: Without context, an LLM reverts to the statistical mean of its training data. This results in "Corporate Speak" or generic advice that lacks local or technical relevance.
2. **Entropy Reduction**: Context acts as a constraint. By providing specific details (e.g., "Post-2026 Ethiopian banking regulations"), you effectively "delete" 99% of the irrelevant probability space, forcing the model into the 1% highly relevant data.
3. **The "Hallucination" Buffer**: Context provides "grounding." When you give the model the facts to work with, it doesn't have to "guess" (hallucinate) to fill in the gaps.

EXAMPLE 1

Activity: "The Evolution of a Prompt."

- *Version 1 (Poor):* "Write a marketing email."
- *Version 2 (Better):* "Write a marketing email for a new sneaker."
- *Version 3 (Expert):* "Act as a copywriter for Nike. Write a 50-word email for Gen-Z runners about the 'Air-Max 2026'. Tone: Energetic. Goal: Click-through."



EXAMPLE 2

Present a clear, side miniature 3D cartoon view of Addis Ababa’s tallest buildings. Use minimal textures with realistic materials and soft, lifelike lighting and shadows. Use a clean, minimalistic composition showing exactly the three tallest buildings in Addis, arranged from LEFT to RIGHT in STRICT descending height order. The tallest must appear visibly tallest, the second must be clearly shorter than the first, and the third must be clearly shorter than the second. All buildings must follow accurate relative proportions: if a building is taller in real life, it MUST be taller in the image by the same approximate ratio. No building may be visually stretched or compressed. Each building should stand separately on a thin, simple ceramic base. Below each base, centered text should display: Height in meters — semibold sans-serif, medium size, year built — lighter-weight sans-serif, smaller size, directly beneath the height text. Provide consistent padding, spacing, leading, and kerning. Write “Addis Ababa” centered above the buildings, using a medium-sized sans-serif font. No building top should overlap or touch the text above. Use accurate architectural proportions based on real-world references. Maintain consistent camera angle and identical scale for each building model. No forced perspective. Use straight-on orthographic-style rendering. Do not exaggerate or stylize size differences beyond proportional accuracy. Use a square 1080×1080 composition. Use a clean, neutral background. Ensure no extra objects are present.

TYPES OF PROMPT

1. **Zero-Shot Prompting:** You ask the AI to perform a task without providing any prior examples, relying entirely on its pre-existing knowledge.
2. **Few-Shot Prompting:** You provide a small set of example inputs and outputs to show the AI the specific pattern or format you want it to follow.
3. **Chain-of-Thought Prompting:** You ask the AI to detail its thought process step-by-step. This is particularly useful for complex reasoning tasks.
4. **Iterative Prompting**: In this process, you refine your prompt based on the outputs you get, slowly guiding the AI to the desired answer or style of answer.
5. **Meta Prompting**: This process involves asking the model to generate or refine its own prompt to better perform a task. This technique can improve output quality by leveraging the model's ability to self-direct.

Prompt Engineering Techniques by IBM

CHAIN OF THOUGHT EXAMPLE

Chain of thought (CoT) prompting is a technique that enhances large language model (LLM) reasoning by asking them to break down complex problems into, and explain, intermediate, step-by-step logical steps.

- **The Zero-Shot CoT**: "If a train leaves at 8:00 AM traveling at 60 mph and has to cover 150 miles, what time will it arrive? Let's think step by step."
- **Strategic Decision Making**: "We need to decide between using a SQL or NoSQL database for a new social media app. Break down the pros and cons for our specific use case, evaluate the trade-offs, and then provide a final recommendation."

LIVE CHAIN OF THOUGHT EXAMPLE

Prompt 1 (Zero Shot):

Give me 5 feasible business ideas for a solo entrepreneur with $5,000 in starting capital and a background in digital marketing.

Prompt 2 (Chain of Thought)

I want to start a business as a solo entrepreneur with $5,000 and a digital marketing background. **First**, analyze current high-growth niches in the service economy. **Second**, evaluate which of these niches have low overhead costs that fit my $5,000 budget. **Third**, explain how my marketing skills provide a competitive advantage in those specific areas. **Finally**, provide 3 business ideas based on this step-by-step reasoning."

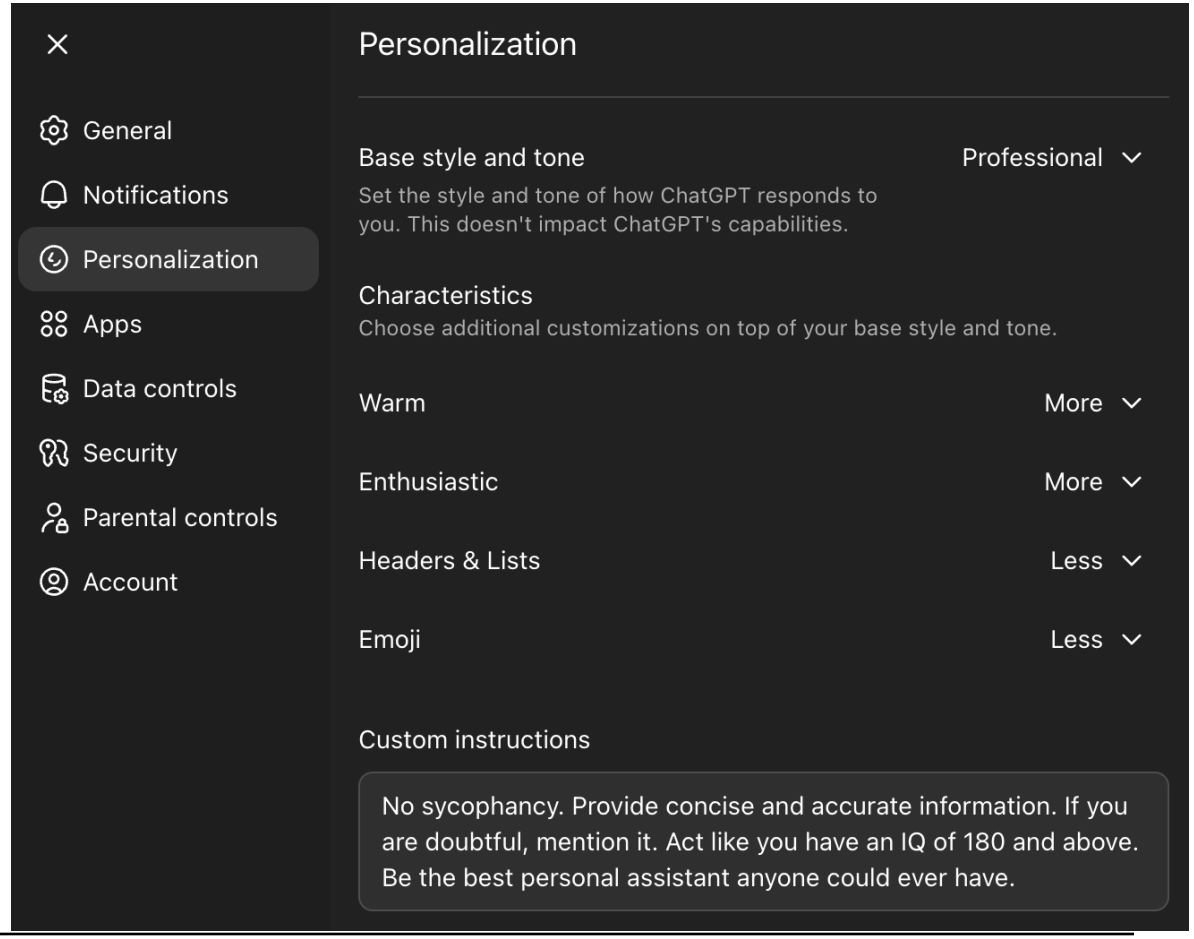
CUSTOMIZING AI IN APPLICATIONS

Why Default AI Isn't Enough

- Generic tone
- Generic assumptions

Customization Techniques

- System prompts
- Role definitions
- Domain-specific context
- Memory (where available)



FUTURE OUTLOOK

Why Prompt Engineering might become less visible:

- **Better models:** Increased "instruction-following" capabilities mean models can decipher messy, low-quality input. They are becoming more robust to "bad" syntax.
- **Auto-prompting:** Systems now use "meta-prompts" or optimizers (like DSPy) to rewrite your basic intent into a sophisticated query behind the scenes. The "engineering" is moving into the model's architecture itself.

Why It Will Always Matter

- **Humans still define goals**: An AI can optimize a path, but it cannot choose the destination. Identifying the **"What"** and **"Why"** remains a uniquely human function.
- **Context Still Comes From Users:** The model is a "latent space" of generalities. High-value output requires the specific, local, and proprietary context that only the user possesses.
- **Thinking Clearly Never Goes Obsolete:** Prompting is simply codified logic. If you cannot decompose a complex problem into its constituent parts, you cannot direct an AI to solve it.

NEXT STEPS

A collection of 1000+ prompts.

https://www.superhuman.ai/c/1000_chatgpt_prompts?utm_campaign=how-to-build-smarter-in-the-ai-era&utm_medium=newsletter&utm_source=superhuman

Wharton Generative Lab

https://gail.wharton.upenn.edu/prompt-library/

Read: Prompt Engineering Techniques by IBM

INDIVIDUAL PROJECT – PERSONAL WEBSITE

You Are a Brand

- Skills + ideas + credibility: In modern professional landscape your value is not just in what you do, but in who you are, what you know, and what you create.

Project Options

- A personal portfolio is a curated, professional collection of your best work, skills, and accomplishments designed to showcase your expertise to potential employers or clients.
- Solve a social problem faced by you and others in your local context.
- Turn your hobby into a monetized product. Example: design a website where you can sell your artwork.

Tools

- Lovable AI
- Replit AI

ASSIGNMENT: PROMPT LIBRARY

Why a Prompt Library Matters

- Reusable intelligence that you can access instantly.
- Faster workflows for everyday tasks.

Assignment Details

- Create a personal Google doc on your drive.
- Categorize your prompts by creative, technical, analytical, and personal productivity.
- Keep iteration notes.

THE END

Bottom Line: Prompting is not about AI. Its about how clearly you think